Department of Oral Medicine and Radiology

Instinctive Reflex – Relearn to Resolve – A Pilot Study

Guided by :

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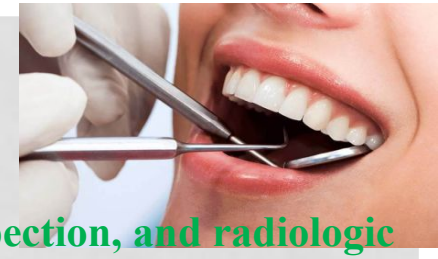
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INTRODUCTION

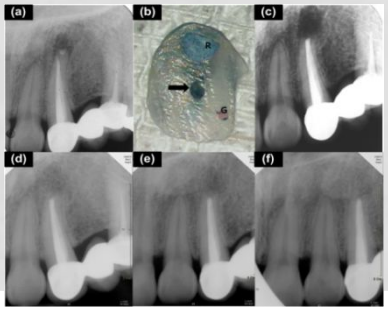


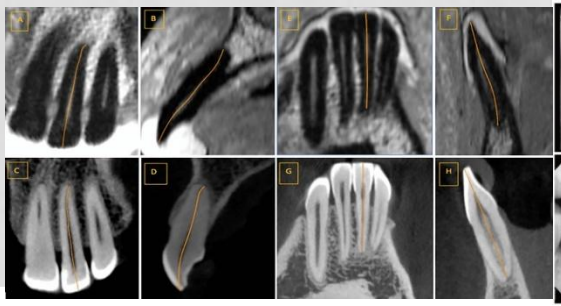
- Oral Diagnosis" literally means "**analysis of the inner mouth".**
- An effective oral diagnosis can be done through **oral inspection, external inspection, and radiologic examination**
- One of the most **important assistive diagnosis** techniques in dentistry is **Radiologic examination**.
- Radiographs used in dental radiology can be divided into two categories:

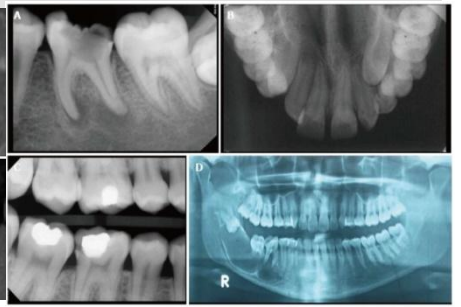
Intraoral and Extra-oral



- ❖ It is usually used to obtain **additional information** beyond that available from a clinical examination or their history which is to is to make a diagnosis and formulate treatment plan.
- ❖ From the simple intra-oral peri apical X-rays, advanced imaging techniques like computed tomography, cone beam computed tomography, magnetic resonance imaging and ultrasound have also found place in modern dentistry.







Types of radiograph for diagnosis

Intra oral radiograph

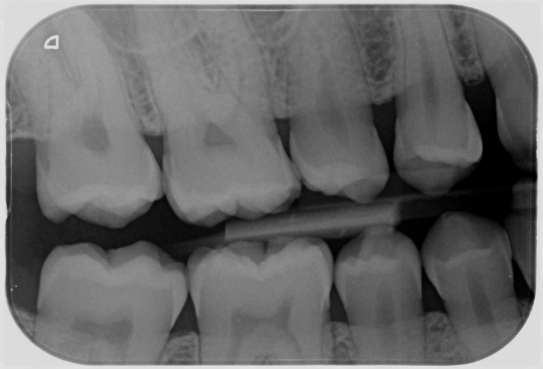
- provides detailed information about the teeth and the surrounding tissues.

IOPA



- Bitewing or inter-proximal radiographs are taken to evaluate inter-proximal surfaces of 3-4 upper and lower teeth simultaneously

Bitewing



- An occlusal radiograph displays a large segment of a dental arch that cannot be viewed on a periapical radiograph, such as a cyst.

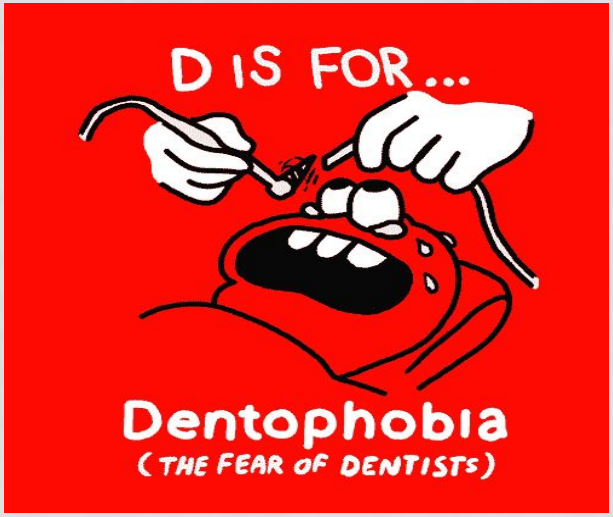
occlusal



ANXIETY AND FEAR RELATED TO DENTAL TREATMENT

- Gagging is a prevalent, intrusive concern in dental care which is associated with dental care-related fear, fear of pain, and **negative perceptions of oral healthcare professionals and dental treatment.**





The purpose of this study was to use an alternative approach on intraoral periapical radiography on gaggers









AIM

- The aim of this study was to **adopt a alternative approach described by (chaudhary et al 2018**) with which the third molar radiograph can be obtained with least discomfort

Methodology



Pt who reported to department of oral medicine and radiology with various complaints



Informed consent and patient is given a gagging questionnarie(50 subjects) which is adopted from Randall et al (2014).



Patient who has complain in third molar (either maxilla or mandible) (30 subjects) were subjected to radiograph with positive gag reflex



STUDY PRO-FORMA

PATIENT NAME : AGE:

OP NO : Gender

Gagging Behaviour Questionnaire

1. **Have you had problems with gagging during dental visits? 0 – No 1 – Yes**

2. **If you have ever had problems with gagging, how severe ave these problems been?**

0 – Very mild 1 – Mild 2 – Neither mild nor severe 3 – Severe 4 – Very severe

If you have ever had problems with gagging during dental visits, how often has gagging interrupted dental treatment?

0 – Never 1 – Rarely 2 – Sometimes 3 – Frequently 4 – Almost always or always

4. **What triggers your gagging during dental visits?**

A. Fingers B. Dental instruments C. Radiographic film D. Impression material E. Other stimuli

5. **Do you have problems with gagging at times other than dental visits?0 – No 1 – Yes**

6. **If you ever had problems with gagging at times other than dental visits, how severe have these problems been?**

0 – Very mild 1 – Mild 2 – Neither mild nor severe 3 – Severe 4 – Very severe

Gagging Behavior Questionnaire

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7. What triggers your gagging at these other times? _____

Reference :
Gagging and Associations with Dental Care-Related Fear, Fear of Pain, and Beliefs about Treatment
Cameron L. Randall, M.S., doctoral candidate, Grant P. Shulman, B.S., graduate student, Richard J. Crout, D.M.D., M.S., Ph.D., Professor Emeritus, and Daniel W. McNeil, Ph.D., Eberly Distinguished Professor
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NIHMSID: NIHMS574564
PMID: 24789238

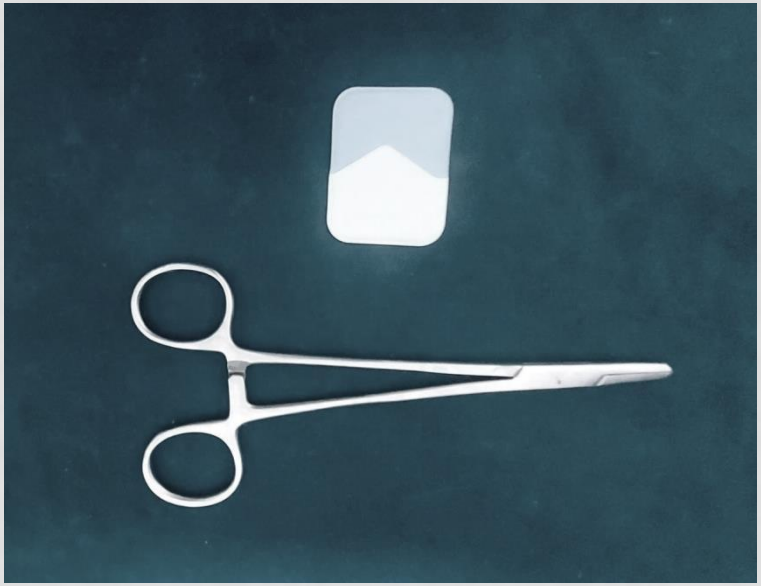
Inclusion criteria

- Patient with positive gag reflex
- Patients who has complain in third molar region
- Patient who are willing to take part in study

Exclusion criteria

- not willing
- Patient who did not having complain in 3 rd molar

Armamentarium



Needle holder

Radiographic film

METHODOLOGY

- The patient is seated upright in the dental chair with the mid-saggital plane perpendicular to the floor for both maxillary and mandibular arches



Exposure parameter :

- **Tube voltage – 60kvp**
- **Tube current – 1 mA**
- **Exposure time - 0.02 s**

2.Film placement :

- An adult size films (31 mm × 41 mm) is inserted in the patient's mouth with the help of a needle holder in a direction parallel to the occlusal plane.
- The film is placed as posteriorly as possible such that it completely covers the occlusal surface of the third molar to be radiographed.
- The patient is advised not to apply excessive pressure while holding the film to prevent damage to the film emulsion



Film placement using needle holder for maxillary right third molar imaging. Note the side of film with emboss dot facing toward the maxillary arch



Film placement using needle holder for mandibular right third molar imaging. Note the side of film with emboss dot facing toward the mandibular arch

3. Projection of xray :

- After placing the film, the patient's head is tilted slightly to the opposite side to bring the X-ray film and the teeth to be radiographed in a desirable relationship with the tube head

- **vertical angulation - +60° to + 65° for maxillary third molar**
- **vertical angulation - −25° to −30° mandibular third molar**

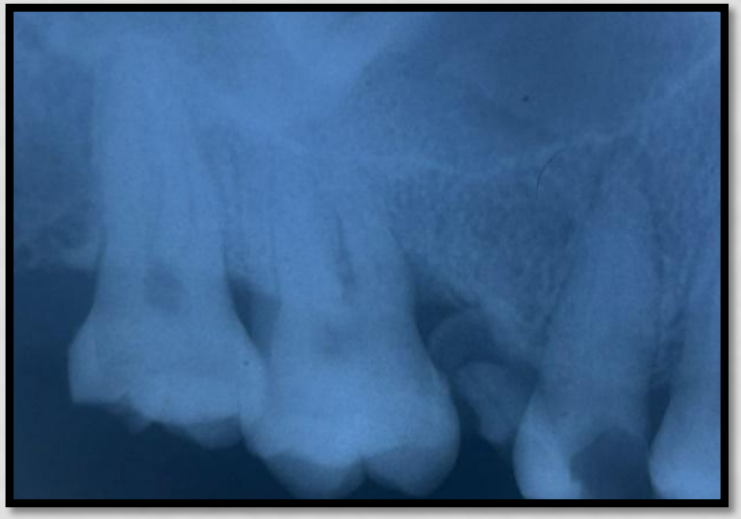


Patient's head position and placement and angulation of the X-ray tube head for right maxillary third molar. Note the tilting of the patient's head to the opposite side

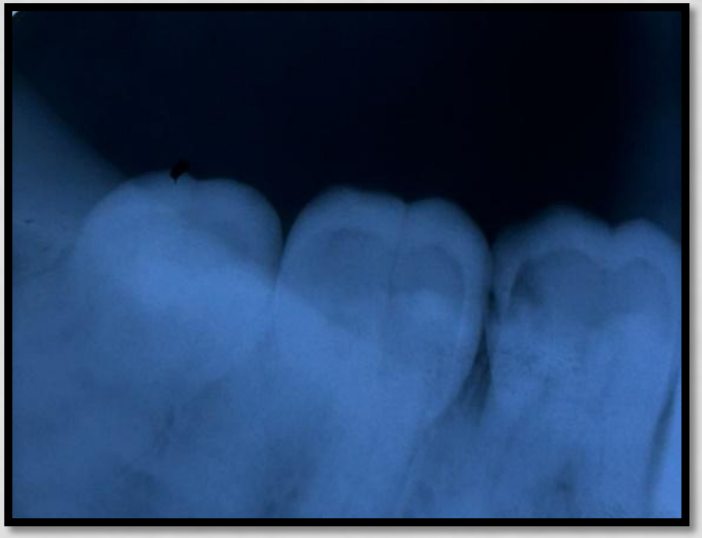


Patient's head position and placement and angulation of the X-ray tube head for right mandibular third molar. Note the tilting of the patient's head to the opposite side

Resultant image



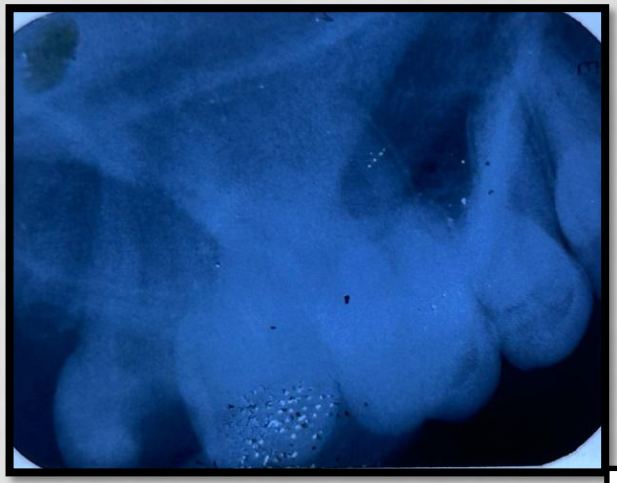
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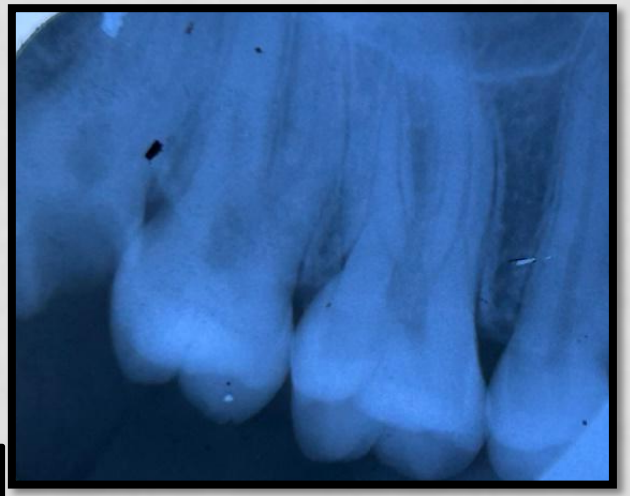


IOPA irt 38

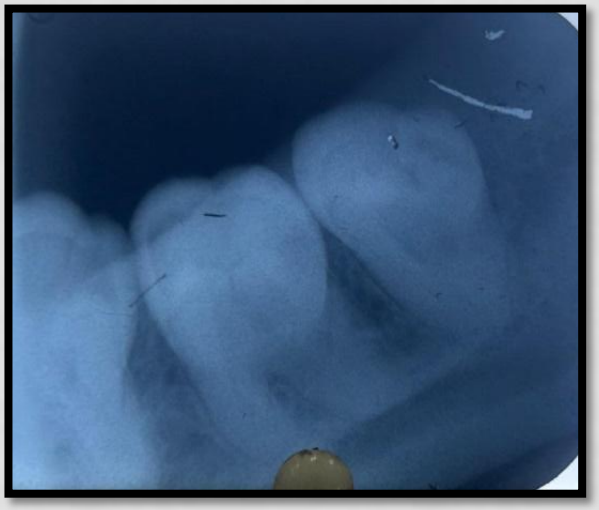
RESULTANT IMAGE VS CONVENTIONAL

IOPA irt 18



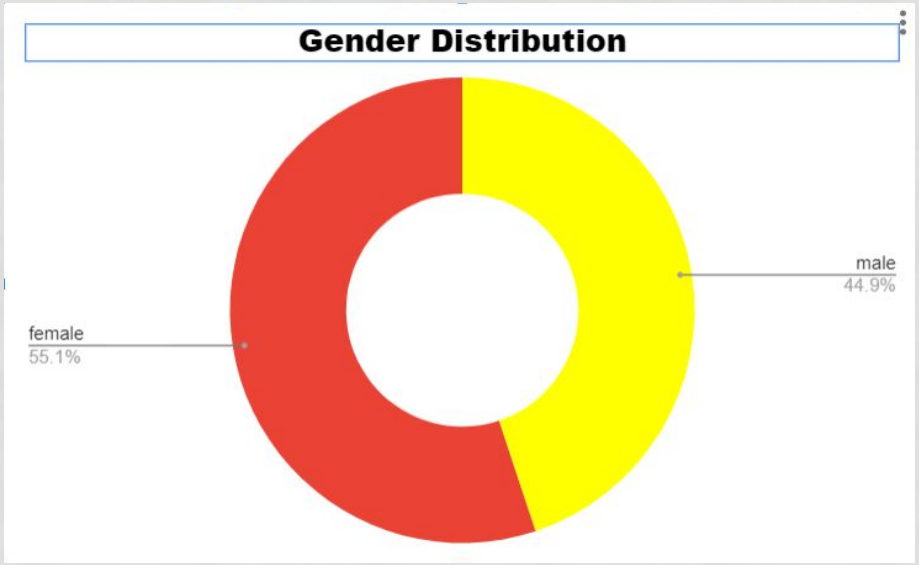


IOPA irt 48



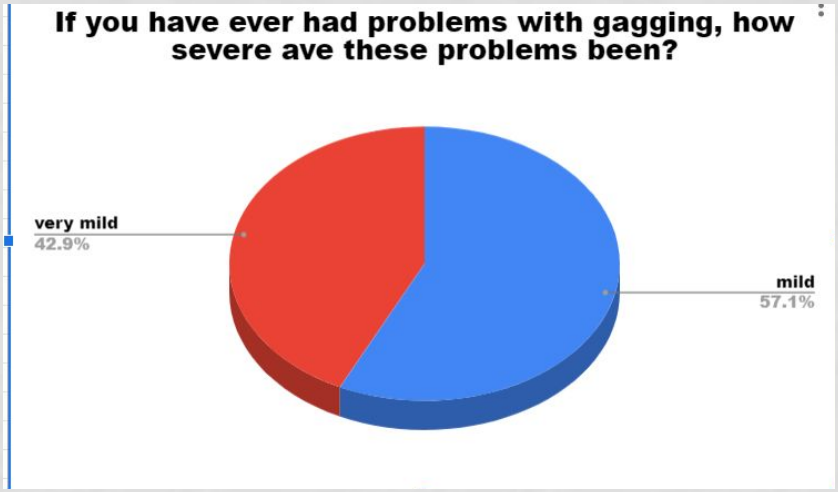


RESULTS



Male

Female

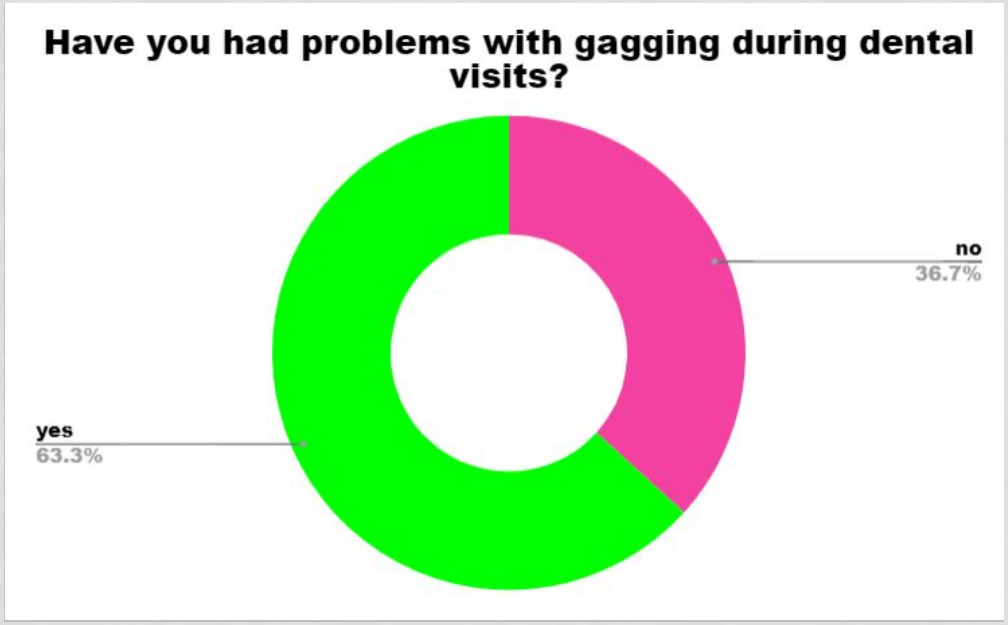


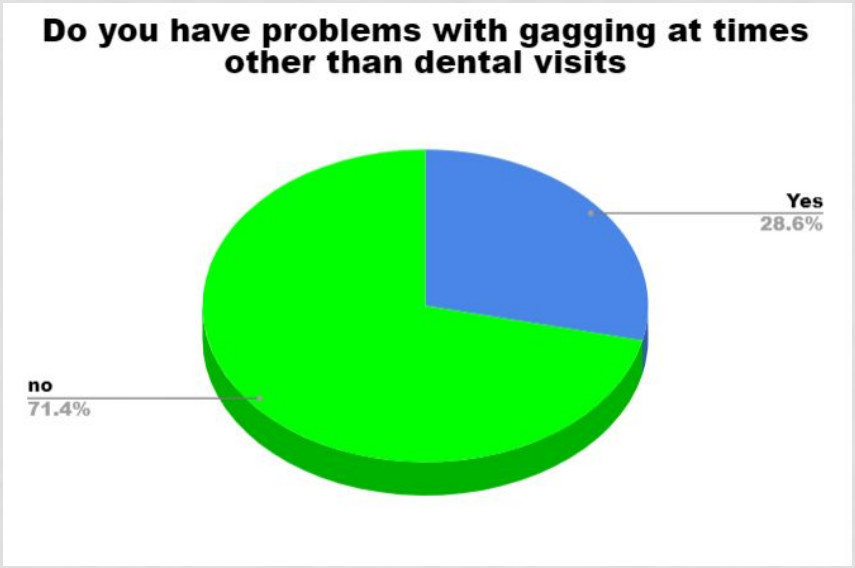
Very mild

Mild

Yes

No



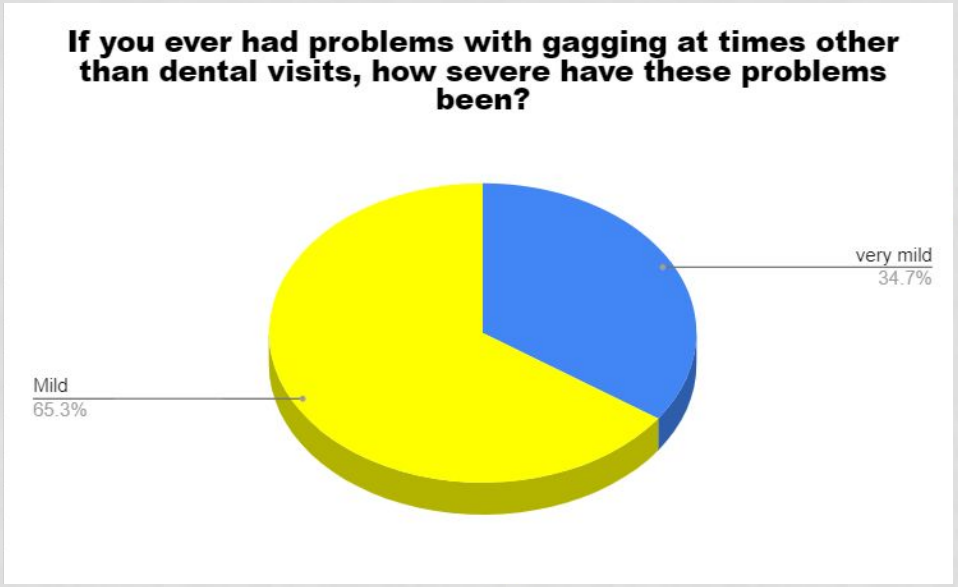


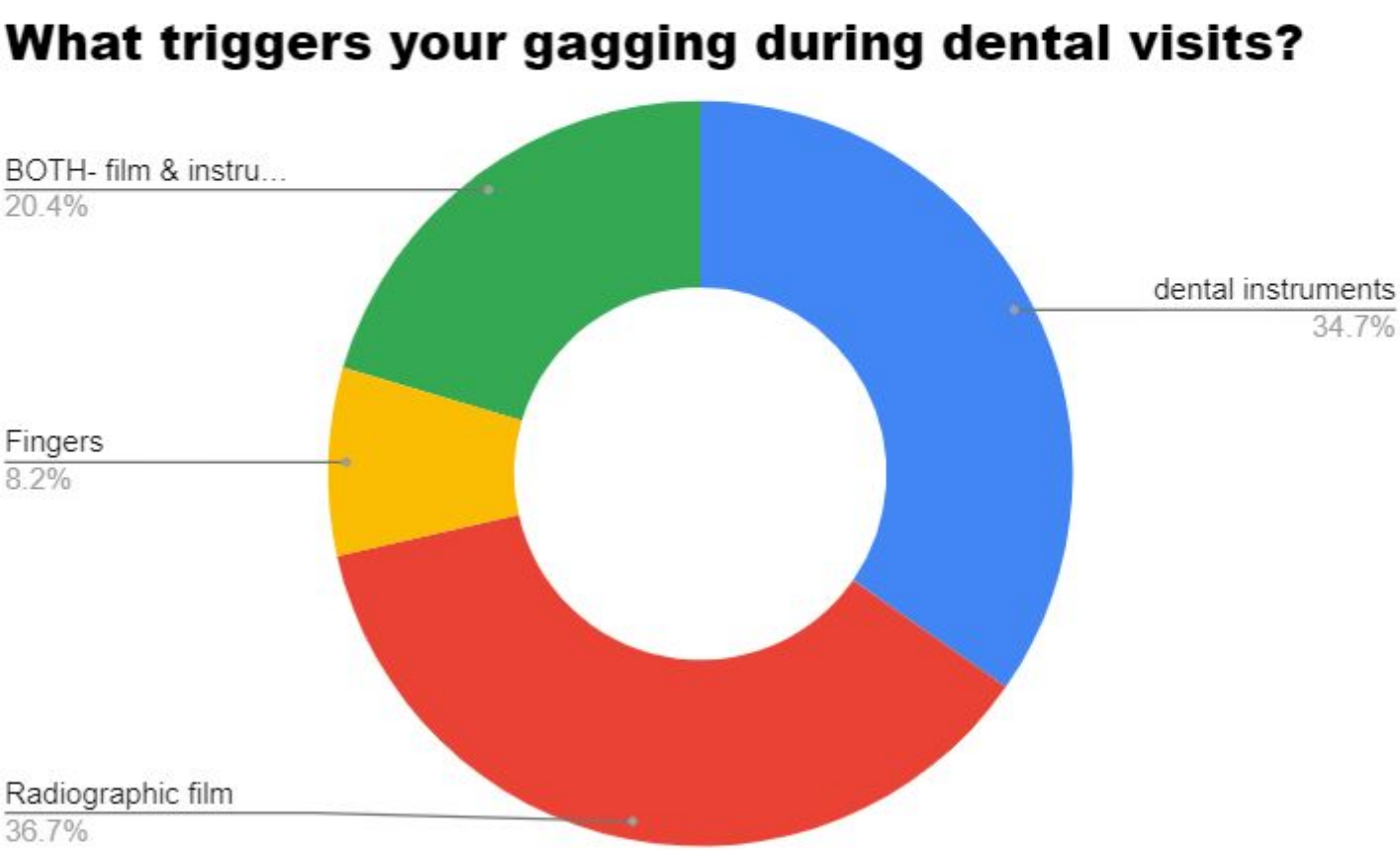
no

Yes

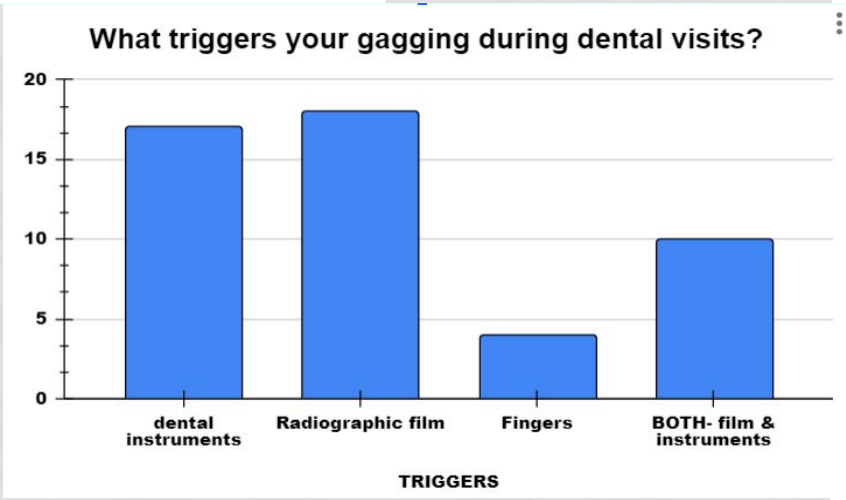
Very mild

Mild





- Radiographic film
- Dental instruments
- Fingers
- Both - Radiographic film and instruments



DISCUSSION

Most commonly faced problems gag reflex even at the slightest provocation.
Gagging third molar imaging using periapical radiographs.
Reason for gagging apprehension and fright of unknown procedures, presence of very sensitive tissues which precipitate gag.

Different measures to overcome this, which includes
1. Brief explanation to the patient radiographic procedure, demonstrating the procedure on other patient which helps - reduce the anxiety of the patient.
2. Another method indulging the patient in some other activities asking the patient to hold breath or keep foot or arm suspended during film placement and exposure which may cause distraction of the patient and thus relieving the gag reaction.
3. In severe cases, topical anesthetic agent may be used.
If none of the above-mentioned measures is able to relieve
Extraoral radiography is advised.
Extraoral comparatively higher doses of radiation but also it increases the cost of radiography.

Advantages

- **Ease of film positioning in patient's mouth, □ more patient cooperation.**
- **Least discomfort**

Limitation of this technique

1. Slight amount of elongation of the resultant image which may occur and Film Bending or damage to the film emulsion due to excessive pressure applied by the patient while holding the film between maxillary and mandibular teeth.

2. Technique sensitive

FUTURE PERSPECTIVES

- **Increased Sample size**
- **Limitation of errors**
- **Alternative approach implementation and comparison**

CONCLUSION

Most commonly advised treatment for third molars is extraction either due to caries or impaction.

This modified technique used since the resultant image shows the full length of the third molar including crown and root morphology relation to surrounding vital structures which are required by the surgeons before extraction.

Diagnosis of INSTINCTIVE REFLEX Through History and clinical examination

The RELEARNING of this technique RESOLVE the gag reflex with least discomfort and aid in diagnosis.

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